

1.

taking input from user

```
user_text = input("Enter some text: ")
```

```
print("Saving your text to file...")
```

writing into file (will create file if not exists)

```
with open("output.txt", "w") as file_obj:
```

```
    file_obj.write(user_text)
```

```
print("File saved successfully!")
```

```
print("\nNow reading the file...")
```

reading from file

```
with open("output.txt", "r") as file_obj:
```

```
    content = file_obj.read()
```

```
print("\nFile content is:")
```

```
print(content)
```

2.

reading file to count details

```
print("Opening sample.txt...")
```

```
with open("sample.txt", "r") as file_obj:
```

```
    content = file_obj.read()
```

```
print("Calculating details...")
```

```
lines = content.split("\n")
```

```
words = content.split()
```

```
chars = len(content)
```

```
print("\nFile Analysis:")
```

```
print("Total Lines:", len(lines))
```

```
print("Total Words:", len(words))
```

```
print("Total Characters:", chars)
```

3.

```
# copying content from source to destination
```

```
print("Reading source file...")
```

```
with open("source.txt", "r") as src_file:
```

```
    data = src_file.read()
```

```
print("Writing to destination file...")
```

```
with open("destination.txt", "w") as dest_file:
```

```
    dest_file.write(data)
```

```
print("File copied successfully!")
```

4.

appending new text to log file

```
new_text = input("Enter text to add in log file: ")
```

```
print("Adding text to log.txt...")
```

```
with open("log.txt", "a") as file_obj:
```

```
    file_obj.write(new_text + "\n")
```

```
print("\nUpdated log file content:")
```

```
with open("log.txt", "r") as file_obj:
```

```
    print(file_obj.read())
```

5.

taking number input and converting to integer

```
user_input = input("Enter a number: ")
```

```
print("Trying to convert...")
```

```
try:
```

```
    num = int(user_input)
```

```
    print("Conversion successful, value is:", num)
```

```
except ValueError:
```

```
    print("Oops! That was not a valid number")
```

6.

opening file entered by user

```
file_name = input("\nEnter file name to read: ")
```

```
print("Opening file...")
```

```
try:
```

```
    with open(file_name, "r") as file_obj:
```

```
        print("\nFile content:")
```

```
        print(file_obj.read())
```

```
except FileNotFoundError:
```

```
    print("File not found, please check the name")
```

7.

division with error handling

```
print("\nLet's perform division")
```

```
try:
```

```
    num1 = int(input("Enter first number: "))
```

```
    num2 = int(input("Enter second number: "))
```

```
    result = num1 / num2
```

```
    print("Result is:", result)
```

```
except ValueError:
```

```
    print("Please enter valid numbers only")
```

```
except ZeroDivisionError:
```

```
    print("Cannot divide by zero, try again")
```

8.

```
# accessing list element using index
```

```
numbers = [10, 20, 30, 40, 50]
```

```
print("\nList is:", numbers)
```

```
try:
```

```
    index = int(input("Enter index to access: "))
```

```
    print("Element at index:", numbers[index])
```

```
except IndexError:
```

```
    print("Invalid index, out of range")
```

9.

```
# creating custom exception
```

```
class NegativeNumberError(Exception):
```

```
    pass
```

```
print("\nChecking for positive number...")
```

```
try:
```

```
    num = int(input("Enter a positive number: "))
```

```
if num < 0:  
    raise NegativeNumberError("Negative number entered!")
```

```
print("Good! You entered:", num)
```

```
except NegativeNumberError as e:  
    print(e)
```

10.

```
# keep asking user until valid division happens
```

```
print("\nTry division until correct input is given")
```

```
while True:
```

```
    try:
```

```
        num1 = int(input("Enter first number: "))
```

```
        num2 = int(input("Enter second number: "))
```

```
        result = num1 / num2
```

```
        print("Final result is:", result)
```

```
        break # exit loop when successful
```

```
except ValueError:
```

```
    print("Please enter numbers only")
```

```
except ZeroDivisionError:
```

```
    print("Cannot divide by zero, try again")
```

11.

demonstrating finally block

```
print("\nTrying to open sample.txt...")
```

```
try:
```

```
    file_obj = open("sample.txt", "r")
```

```
    print("\nFile content:")
```

```
    print(file_obj.read())
```

```
except FileNotFoundError:
```

```
    print("File not found!")
```

```
finally:
```

```
    print("Program execution completed (finally block)")
```

12.

function to add two values

```
def add_values(a, b):
```

```
    try:
```

```
        print("Sum is:", a + b)
```

```
    except TypeError:
```

```
        print("Invalid input! Only numbers allowed")
```

```
print("\nTesting add function")
```

```
add_values(10, 20)    # valid
add_values(10, "abc") # invalid test
```

13.

keep asking user until valid integer is entered

```
print("\nEnter a valid integer")
```

```
while True:
```

```
    try:
```

```
        user_num = int(input("Enter integer: "))
```

```
        print("You entered:", user_num)
```

```
        break
```

```
    except ValueError:
```

```
        print("Not an integer, try again")
```